COMPLETE SETUP GUIDE / OCTOBER 2026

Raspberry Pi 4 Headless Remote Access

Fresh Raspberry Pi OS setup with SSH, RustDesk desktop access, and Tailscale private networking - all without a monitor, keyboard, or mouse on the Pi.

> **Assumes:** Raspberry Pi 4 Model B, Raspberry Pi OS 64-bit Desktop (Trixie), a Windows 10/11 computer, and a microSD card or compatible SSD.

Component	Selected setup
Device	Raspberry Pi 4 Model B
Pi hostname / username	`mypi4` / `piadmin` (examples)
Operating system	Raspberry Pi OS (64-bit) Desktop - not Lite
First connection	SSH from Windows PowerShell
Remote desktop	RustDesk ARM64 with a permanent password
Private remote networking	Tailscale on both the Pi and Windows

> **Back up first.** Imaging will erase the selected card or drive. Use a reliable Pi 4 power supply. Keep SSH working even after RustDesk works.

1 Prepare the installation media on Windows

1. Install **Raspberry Pi Imager** from raspberrypi.com/software.
2. Connect your microSD card or SSD to the Windows PC.
3. In Imager choose **Raspberry Pi 4** > **Raspberry Pi OS (64-bit), Desktop** > the correct storage.

2 Set the Pi's name, password, Wi-Fi and SSH

In the Imager customization screens (labels can vary), use these values:

Setting	What to choose
Hostname	`mypi4`
Username	`piadmin`
Pi login password	Create a strong password and save it privately
Wi-Fi	Your SSID, Wi-Fi password and correct Wi-Fi country
SSH	Enable SSH. Public-key authentication preferred; password login is OK for initial setup.

3 Boot the Pi 4 and connect from Windows

Insert the card in the Pi. Connect Ethernet (if available) or rely on the configured Wi-Fi; then supply power. No display or input devices are needed.

On Windows, open **PowerShell** and run:

```
ssh piadmin@mypi4.local
```

If the `.local` name is not found, look up the Pi's address in your router's device list and use the real IP address instead:

```
ssh piadmin@192.168.1.100
```

Accept the SSH fingerprint if expected, then enter your **Pi login password** (unless using a key). Once connected:

```
cat /proc/device-tree/model
cat /etc/os-release
uname -m
```

Confirm Raspberry Pi 4, Raspberry Pi OS and `aarch64`.

4 Update Raspberry Pi OS

```
sudo apt update
sudo apt full-upgrade -y
sudo reboot
```

Wait for restart and reconnect to SSH. A DHCP reservation in your router can help keep its LAN address predictable.

5 Start the X11 desktop automatically

RustDesk's Wayland access is still experimental; an X11 login session is a more predictable choice for this headless setup.

```
sudo raspi-config
```

1. Choose **Advanced Options > Wayland > W1 X11**.
2. Choose **System Options > Boot / Auto Login > Desktop Auto Login**.
3. Finish; you can defer reboot until after Step 6.

> **Local access caution:** Desktop auto-login means someone with physical access to a running Pi may use its desktop without signing in. SSH and RustDesk still have separate credentials.

6 Force a display with no HDMI monitor attached

Back up the boot command line and edit it:

```
sudo cp /boot/firmware/cmdline.txt \
  /boot/firmware/cmdline.txt.bak
sudo nano /boot/firmware/cmdline.txt
```

At the end of the file's **existing single line**, add one space and this parameter:

```
video=HDMI-A-1:1920x1080M@60D
```

Save with **Ctrl+O**, Enter, then **Ctrl+X**. Do not insert a new line. Reboot:

```
sudo reboot
```

After reconnecting through SSH, check:

```
cat /proc/cmdline
pgrep -a Xorg
systemctl status lightdm --no-pager
```

> **Headless compatibility:** Forcing HDMI is a practical workaround, not a guarantee. Some Pi 4 and OS combinations also need an HDMI dummy plug or suitable synthetic EDID. If `pgrep` returns nothing, troubleshoot before continuing.

7 Install RustDesk (ARM64 build)

The Raspberry Pi 4 with 64-bit OS uses the RustDesk **aarch64** Debian package. These commands use stable version 1.4.9:

```
VERSION=1.4.9
URL=https://github.com/rustdesk/rustdesk/releases
wget -O /tmp/rustdesk-arm64.deb \
  "$URL/download/$VERSION/rustdesk-$VERSION-aarch64.deb"
```

```
sudo apt install -y /tmp/rustdesk-arm64.deb
sudo systemctl enable --now rustdesk
systemctl status rustdesk --no-pager
```

The service should be **active (running)**. If the release changes, get the current ARM64 `.deb` from github.com/rustdesk/rustdesk/releases.

8 Create your permanent RustDesk password

> **You choose this password yourself.** It is **not** your Raspberry Pi login/SSH password. You will use it on Windows when connecting through RustDesk.

8A - Make sure you are connected to the Pi over SSH

```
ssh piadmin@mypi4.local
```

8B - Copy all four lines into the SSH terminal

```
read -rsp "Enter new RustDesk password: " RD_PASS
echo
sudo rustdesk --password "$RD_PASS"
unset RD_PASS
```

1. The terminal displays: **Enter new RustDesk password:**
2. Think of a **long, unique password** and type it. No characters or asterisks appear. **That is normal.**
3. Press **Enter**. If sudo prompts for a password, enter the **Pi login password**.
4. The other commands set the RustDesk password and remove it from the temporary shell variable.
5. Save your new RustDesk password in a password manager. **Never share it in chat.**

These commands must be run inside your Raspberry Pi SSH session (Linux shell), not directly as PowerShell commands on Windows.

8C - Get the RustDesk device ID

```
sudo rustdesk --get-id
```

Record the ID separately from your password. If a connection later rejects the password, check that the service is active and repeat Step 8B.

9 Test RustDesk from Windows

1. Download and install RustDesk on Windows from rustdesk.com.
2. Enter your Pi's RustDesk device ID and choose **Connect**.
3. When prompted, enter the **RustDesk password** you created in Step 8.
4. Check that the Pi desktop displays correctly and accepts clicks/typing.

```
sudo reboot
```

Once it comes back, reconnect using RustDesk to confirm unattended startup.

NEW CHAPTER: TAILSCALE ON YOUR PI

ADDED: SECURE PRIVATE NETWORK ACCESS

10. Install Tailscale for remote access from anywhere

Why add Tailscale? It gives your Pi and Windows computer a private, encrypted network even when they're in different locations. It is especially useful for reaching SSH without router port forwarding.

> **RustDesk vs. Tailscale:** RustDesk shows the desktop. Tailscale connects devices at the network level. Installing Tailscale alone **does not** route RustDesk's ID-based sessions through Tailscale.

10A - Install Tailscale on Raspberry Pi 4 over SSH

Keep the SSH terminal connected to your Pi. Install the official Tailscale package using its installer:

```
curl -fsSL https://tailscale.com/install.sh | sh
```

This command downloads and runs the official Tailscale installation script. If you prefer, use Tailscale's manual package instructions linked below instead.

10B - Sign the Pi into your Tailscale network

```
sudo tailscale up
```

You will receive a browser sign-in link, usually beginning with:

```
https://login.tailscale.com/a/...
```

1. Copy the **complete link** from your SSH terminal.
2. Paste it into a browser on your **Windows** computer.
3. Sign into your Tailscale account and approve adding the Pi.
4. Return to the Pi's SSH terminal and verify the status:

```
tailscale status
tailscale ip -4
systemctl is-active tailscaled
```

`tailscale ip -4` normally displays an address such as `100.101.102.103`. **This is only an example;** write down your Pi's actual Tailscale IP.

Command	What it tells you
`tailscale status`	Whether the Pi is connected to your tailnet
`tailscale ip -4`	The Pi's private Tailscale IPv4 address
`systemctl is-active tailscaled`	Whether Tailscale's background service is active

Official Linux installation: tailscale.com/docs/install/linux

10. Tailscale, continued

10C - Install Tailscale on your Windows computer

1. Visit tailscale.com/download/windows.
2. Download and run the Windows installer.
3. Find the Tailscale icon in the **system tray** (near the clock; expand hidden icons if needed).
4. Choose **Log in** and use the **same Tailscale account/tailnet** as your Raspberry Pi.
5. Check that both your Windows computer and `mypi4` appear in the device list at login.tailscale.com/admin/machines.

10D - Connect to your Raspberry Pi using Tailscale

Disconnect from your local Wi-Fi or test from another network if practical (e.g., a mobile hotspot). Open **PowerShell** on Windows:

```
ssh piadmin@100.101.102.103
```

Replace `100.101.102.103` with the actual IP from `tailscale ip -4`. The Pi still asks for its usual SSH user password or SSH key - not its RustDesk password.

With MagicDNS enabled, this may work too:

```
ssh piadmin@mypi4
```

> **Success test:** You can log into the Pi via SSH from your Windows PC even though they are on different networks. Keep Tailscale and SSH installed as a recovery route if RustDesk ever shows a black screen.

10E - What to do after reboot

Tailscale normally starts automatically after installation. Check from an SSH session:

```
systemctl is-active tailscaled
tailscale status
```

> **Security:** Do not forward SSH port 22 or a RustDesk port on your internet router just to use Tailscale. Keep both devices updated, protect your Tailscale account with MFA, and verify remote access after a full power cycle.

Official Windows instructions: tailscale.com/docs/install/windows | Direct device connections: tailscale.com/docs/how-to/connect-to-devices

11. How RustDesk and Tailscale work together

Recommended simple setup: Use the RustDesk **device ID + RustDesk password** for the remote graphical desktop (Step 9). Use the Pi's **Tailscale IP + SSH login** for private command-line administration (Step 10).

Optional advanced method: After RustDesk is working, open RustDesk's settings on the Pi and enable **Security > Enable direct IP access**. The usual direct-access port is `21118`. From Windows with Tailscale connected, enter the Pi's **actual Tailscale IP** (and port if needed) into RustDesk. Test this before relying on it; direct-IP access is **off by default** and may be restricted by firewall rules.

See official RustDesk setting `direct-server` / `direct-access-port`: RustDesk advanced settings.

Final checklist

- [ ] Pi 4 boots without a display or keyboard attached.
- [ ] SSH works over the local network after restart.
- [ ] RustDesk service and X11 desktop are active.
- [ ] RustDesk device ID and permanent password open the desktop on Windows.
- [ ] Tailscale shows both Windows and Pi as online in the same tailnet.
- [ ] SSH via the Pi's `100.x.y.z` Tailscale IP succeeds from another network.
- [ ] Everything still works after switching the Pi off and on again.

Quick troubleshooting over SSH

```
systemctl is-active ssh
systemctl is-active rustdesk
systemctl is-active tailscaled
pgrep -a Xorg
sudo rustdesk --get-id
sudo journalctl -u rustdesk -b --no-pager -n 40
```

Symptom	Where to look first
No Pi access	Power, correct Wi-Fi, router IP, SSH configuration
RustDesk black screen	X11, LightDM, auto-login, HDMI/EDID
Tailscale IP does not connect	Both devices logged into same tailnet; Tailscale status; SSH service

Windows: if the PDF will only preview, not download

In the preview choose the download arrow, or open the PDF in a browser tab and try **Ctrl+S**. Another method is **Ctrl+P > Microsoft Print to PDF**. A ZIP and HTML copy of this guide are also supplied.

Checked 18 October 2026. Commands and screens can change with future OS releases. RustDesk headless support can depend on your exact display/desktop setup.